

Course Code: CSE4005	Course Title: Data Warehousing and Data Mining	TPC	3	2	4
Version No.	1.1				
Course Pre-requisites/ Co-requisites	CSE1006				
Anti-requisites (if any).	None				
Objectives:	1. To introduce concepts and techniques of data warehousing. 2. To edify the underlying concepts and architecture of data mining. 3. To make students understand association mining and cluster analysis.				
COs Mapping with POs and PEOs					
Course Outcomes	Course Outcome Statement			POs / PEOs	
CO1	Understand the functionality of the various data warehousing components.			PO1, PO5	
CO2	Analyse the strengths and limitations of various data mining and data warehousing models.			PO2, PO5	
CO3	Basic Design, implement and evaluate a system using mining principles.			PO2, PO3, PO5	
CO4	Identify, formulate and solve engineering problems using association mining and clustering.			PO2, PO3, PO4, PO5	
TOTAL HOURS OF INSTRUCTIONS: 45					
Module No. 1	Introduction To Data Warehousing				6 Hours
Data preprocessing- Data cleaning – Data integration – Transformation-Data Reduction-Data Warehouse Characteristics- A multidimensional Data Model – OLAP Operations-Types of OLAP server-Data Warehouse Architecture and Implementation.					
Module No. 2	Data Mining Concepts				7 Hours
Introduction to Data Mining – Data Mining Functionalities – Classification of DataMining systems, Data Mining Task primitives-Integration of data mining with database-Major issues in Data mining.					
Module No. 3	Frequent Pattern Mining				8 Hours
Basic concepts – market basket analysis - Efficient and Scalable Frequent Item setMining Methods-Various Kinds of Association Rules- Association Mining to Correlation Analysis					
Module No. 4	Classification And Prediction				8 Hours
Classification - issues regarding classification and prediction -Decision Tree Induction-Bayesian classification - Rule-Based Classification- Associative Classification Prediction: linear – nonlinear regression- Accuracy and Error Measures.					
Module No 5	Cluster Analysis				8 Hours
Types of Data in cluster analysis - Major clustering methods- Partitioning Methods, Hierarchical Methods, Density Based Methods					
Module No 6	Outlier Analysis				8 Hours
Outlier Analysis- Distance-Based Outlier Detection- Density-Based Local Outlier Detection- Applications of Data mining.					
Text Book					
1. Han J. &Kamber, M, “ Data Mining: Concepts and Techniques”, Morgan Kaufmann, Third Edition, 2011.					

References

1. Mehmed Kantardzic "Data Mining : Concepts, Models, Methods and Algorithms" Wiley, Second Edition, 2017
2. Charu C Agarwal, "Data Mining " The Text Book, Springer, 2015
3. Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, Vipin Kumar "Introduction To Data Mining", Pearson Education, 2013
4. Gerardus Blokdyk, "Data warehouse Complete Self-Assessment Guide", 5starcooks, 2018

Course Type	Embedded Theory and Project (ETP)		
Mode of Evaluation	Theory –		75%
	Continuous Assessment Test-1	15%	
	Continuous Assessment Test-2	15%	
	Digital Assignments/Quizzes (Min)	30%	
	Final Assessment Test	40%	
	Project –		25%
Recommended by the Board of Studies on	14th BoS, 11.05.2024		
Date of Approval by the Academic Council	12th Academic Council, 25.05.2024		